

Basic Overview

- **Company:** Synchron
 - **Sector:** Neurotechnology / Brain-Computer Interfaces (BCI) / Medical Devices
 - **Founded:** 2016
 - **Founders:**
 - Thomas Oxley
 - Nicholas Opie
 - Originated from research at the University of Melbourne
 - Goal: Help people with severe paralysis control computers and devices using only their thoughts
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Problem They Solve

- Patients with ALS, stroke, or spinal cord injuries may lose the ability to:
 - Speak
 - Type
 - Use computers
 - Control smart devices
 - Existing BCIs often require invasive brain surgery
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Solution

- Developed a brain-computer interface called the **Stentrode**
- Allows users to control digital devices using brain signals
- Less invasive than traditional BCIs

Main Goal

- Restore communication and digital independence for paralyzed patients
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The Core Technology: Stentrode

What Makes It Different?

- No need to open the skull
- Implanted through blood vessels

How It Is Implanted

1. Catheter inserted through the jugular vein in the neck
2. Device guided through blood vessels
3. Placed in the superior sagittal sinus near the motor cortex
4. Expands like a cardiovascular stent

Device Structure

- Similar to a heart stent
 - Contains tiny electrodes that record brain activity
 - Sits inside a blood vessel rather than inside brain tissue
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System Setup

Internal Components

- Stentrode implanted in blood vessel
- Connected to a transmitter implanted in the chest
- Similar setup to a pacemaker

External Components

- External receiver placed over chest
- Receives brain signals wirelessly
- Sends data to:
 - Computer
 - Smartphone
 - Other digital devices

Result

- Thoughts become digital commands
 - Users can:
 - Move a cursor
 - Click
 - Type
 - Control smart-home devices
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Clinical Trials

Australian Trial (SWITCH)

- Demonstrated long-term safety

- Device remained stable for years
- No major blood vessel blockages
- No major safety concerns

U.S. COMMAND Trial

- Conducted under FDA approval
- Sites included:
 - Mount Sinai Health System
 - UPMC
 - University at Buffalo

Results

Patients successfully:

- Sent texts
 - Wrote emails
 - Managed finances
 - Controlled smart-home devices
 - Communicated independently
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AI Integration

- Added generative AI tools to the platform
- Uses:
 - Predictive text
 - Smart replies
 - Context-aware suggestions

Benefits

- Faster communication
 - Less typing required
 - More natural conversations
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Regulatory Progress

- First BCI company to receive FDA Investigational Device Exemption (IDE) approval for permanent implant trials
- Focused heavily on safety and regulatory approval
- Considered one of the leaders in BCI commercialization

Funding & Investors

Total Funding

- \$145M+ raised

Major Investors

- ARCH Venture Partners
 - Bill Gates (through Gates Frontier)
 - Jeff Bezos (through Bezos Expeditions)
 - Reliance Industries
 - Khosla Ventures
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Competitive Advantage (Moat)

1. Uses Existing Medical Infrastructure

- Similar procedure to cardiovascular stenting
- Doctors already understand the technique
- Easier regulatory pathway

2. Strong Safety Profile

- No direct brain penetration
- Lower surgical risk
- Less damage to brain tissue

3. Reduced Foreign-Body Response

- Avoids glial scarring issues common in implanted electrodes
 - Potentially more stable long-term performance
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Main Weakness

Lower Signal Resolution

Because the electrodes sit inside a blood vessel:

- Signals are weaker than Neuralink's

- Cannot easily record individual neuron activity
- May limit advanced applications

Possible limitation:

- High-precision robotic prosthetic control
 - Complex multidimensional motor tasks
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Key Lessons / Takeaways

1. Innovation Doesn't Always Mean More Complexity

- Synchron improved BCIs by simplifying implantation
- Used existing medical procedures rather than creating new ones

2. Safety Can Be a Competitive Advantage

- Easier approval process
- Greater physician acceptance
- Lower patient risk

3. Infrastructure Matters

- Leverages hospitals' existing catheterization capabilities
- Easier scaling than highly specialized surgeries

4. Different Strategy Than Neuralink

- Neuralink = highest performance
 - Synchron = safest and easiest path to commercialization
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Quick TL;DR

- Brain-computer interface company helping paralyzed patients use technology with their thoughts.
- Uses the **Stentrode**, a stent-like device implanted through blood vessels instead of open brain surgery.
- First BCI company to receive FDA approval for permanent implant clinical trials.
- Raised \$145M+ and backed by investors connected to Bill Gates and Jeff Bezos.
- Biggest advantage is safety and ease of implantation.
- Biggest challenge is lower signal quality compared to Neuralink.

- Excellent example of a startup winning through **practical engineering and regulatory strategy rather than pure technological power.**

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